Dhruv Sharma

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Education

- **University of Cambridge** — Cambridge, United Kingdom
  MPhil in Scientific Computing — *Oct 2025 – Sep 2026*
  Awarded a merit-based scholarship covering full tuition by the Department of Physics.
  Research focus on machine learning and high-performance computing for atomistic simulation, using OpenMP, MPI and CUDA.
  Teaching Assistant for the NST IA Scientific Computing course, teaching Python, numerical computation, and data visualisation.
- **University College London** — London, United Kingdom
  BSc Chemistry with Mathematics; Grade: First-Class Honours — *Sep 2022 – Jun 2025*
  Courses: *Algorithms and Data Structures, Scientific Programming, Computational Chemistry, Mathematics for Physics and Astronomy*
  Activities and Societies: *Quant Society, Asset Management Society*

Experience

- **Cartivo** — London, United Kingdom
  Co-founder — *May 2026 – Present*
  - Built a multi-tenant B2B SaaS platform using a Medusa backend and tenant-isolated database architecture.
  - Implemented customer onboarding, order approvals, payments and operational dashboards for B2B wholesale workflows.
  - Built a tenant control panel to monitor health, gate releases and roll out tested updates across customers.
  - Own product, engineering, deployment and operations end-to-end for a live B2B platform with paying UK customers.
- **Cisco ThousandEyes** — London, United Kingdom
  Software Engineer Intern — *Jun 2025 – Sep 2025*
  - Developed an ARKit/SwiftUI indoor-navigation iOS prototype with a Wi-Fi heatmap to find network dead zones.
  - Built Wi-Fi booster-placement logic with greedy/simulated annealing algorithms and background multithreading.
  - Feature prototype was presented to 1000+ employees at company-wide kickoff, earning recognition from leadership.
  - Integrated C++ network tests into Android & iOS apps to measure service speeds from user devices through their router.
  - Built a cross-platform CMake/Bash pipeline to compile core C++ libraries (OpenSSL, cURL) for Android & iOS.
  - Worked with embedded C++ and mobile Kotlin/Swift teams, fixing complex C++ and CMake build errors.
- **Microsoft** — Edinburgh, United Kingdom
  Software Engineer Intern, Azure for Operators — *Jul 2024 – Sep 2024*
  - Used Ceph to manage data replication between voicemail servers scaling to millions of users.
  - Wrote production code to support zero-downtime server upgrades for customers with five nines uptime.
  - Deployed Linux VMs, automated processes with bash scripts, monitored and managed network services/APIs.
  - Replaced Ansible workflows with Python/Bash automation, improving operational command speed by up to 1000x.
  - Developed server-side code in a large production codebase, managing endpoint selection for a multithreaded service.
  - Developed a Microsoft 365 Copilot extension for researchers to accurately discover papers via the Semantic Scholar API.
- **S-Cube** — London, United Kingdom
  Software Developer Intern — *Jun 2023 – Aug 2023*
  - Applied automatic differentiation and Fourier transforms with NumPy to signal-processing workflows on seismic data.
  - Vectorised zero-lag cross-correlation calculations to improve compute time on numerical data-processing tasks.
  - Automated data extraction from documents using Python with pandas, itables and AWS CLI.
  - Developed chatbots with LLMs, Flask and PostgreSQL, integrated intent detection for API data retrieval.
  - Deployed live web applications with AWS EC2, AWS ELB, Docker, Kubernetes, Nginx, and Git for version control.

Open Source

- **CP2K:** Optimised ML inference from super-linear to near-linear runtime in a Fortran molecular-dynamics codebase. Optimised CUDA GPU implementation and developed MACE graph neural network workflows for fine-tuning and distilling domain-specific atomistic ML models.
- **Movement:** Worked with Sainsbury Wellcome Centre and Gatsby Unit to develop an open-source Python package for animal-movement kinematic analysis in behavioural neuroscience. Coauthored work presented at Measuring Behavior 2024.

Additional Information

- **Skills:** C++, Python, CUDA, MPI, OpenMP, CMake, Fortran, C, Kotlin, Swift, Java, Bash and SQL.
- **Interests:** Brazilian Jiu-Jitsu, Mixed Martial Arts, Weight Lifting, Scuba Diving, Surfing.